

Finite-Ohnesorge validation of drop coalescence

CoMPhy Lab

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Finite inertia slows the coalescing neck before the neck reaches the expected inertial crossover. At $\text{Oh} = 0.6$, the present finite-element method (FEM) reproduces the developed velocity curve of Anthony, Harris and Basaran [1], while its velocity lies below the present Stokes solution by 6.46% at $R_{\min} = 10^{-5}$ and 12.15% at $R_{\min} = 0.03$. We keep comparison with published data separate from numerical verification and convergence. An independent Stokes boundary-integral method (BIM) reproduces the FEM startup where the published Stokes markers differ.

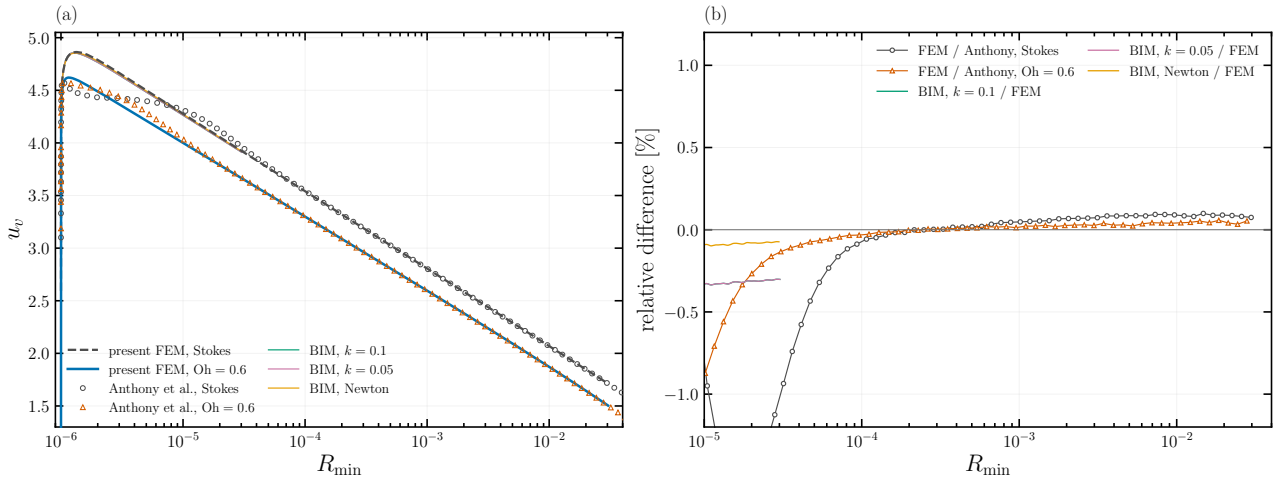


Figure 1: Neck velocity in visco-capillary units. Blue: present FEM at $\text{Oh} = 0.6$; dashed grey: Stokes FEM. Triangles and circles: Anthony et al.’s finite-Oh and Stokes markers. Green, pink and yellow: independent Stokes BIM. Panel (b) compares FEM/Anthony and BIM/FEM minus one in per cent at equal radius, without fitted offsets; early startup is shown separately in Figure 4.

1 Problem and scales

Two equal drops of radius R_d coalesce in a passive exterior of zero density and viscosity. Length, velocity and time scales are R_d , γ/μ and $\mu R_d/\gamma$. With $\text{Oh} = \mu/\sqrt{\rho\gamma R_d}$, the liquid obeys

$$\text{Oh}^{-2}(\partial_t \mathbf{u} + \mathbf{u} \cdot \nabla \mathbf{u}) = \nabla \cdot \boldsymbol{\sigma}, \quad \nabla \cdot \mathbf{u} = 0, \quad \boldsymbol{\sigma} = -p\mathbf{I} + \nabla \mathbf{u} + \nabla \mathbf{u}^T. \quad (1)$$

The initial bridge has $R_0 = 10^{-6}$ and $Z_0 = R_0^2/2$; its meridian $[r - (R_0 + Z_0)]^2 + z^2 = Z_0^2$ joins the spherical drop tangentially. The finite-Oh drops start from rest. The FEM uses Taylor–Hood triangles, tip-graded ALE coordinates and second-order backward differentiation.

The solver time t_v is already visco-capillary. We construct $\tau_v = t_v + t_{\text{con},v}$ by fitting $R_{\min} = A(t_v + t_{\text{con},v})^n$ over $10R_0 \leq R_{\min} \leq 100R_0$, following the contact-time extrapolation of Ref. [1]. The contact shifts are 1.5202×10^{-7} at $\text{Oh} = 0.6$ and 1.4309×10^{-7} in Stokes flow. No additional division by Oh is made, and no offset is fitted to the published markers.

2 FEM against Anthony et al.

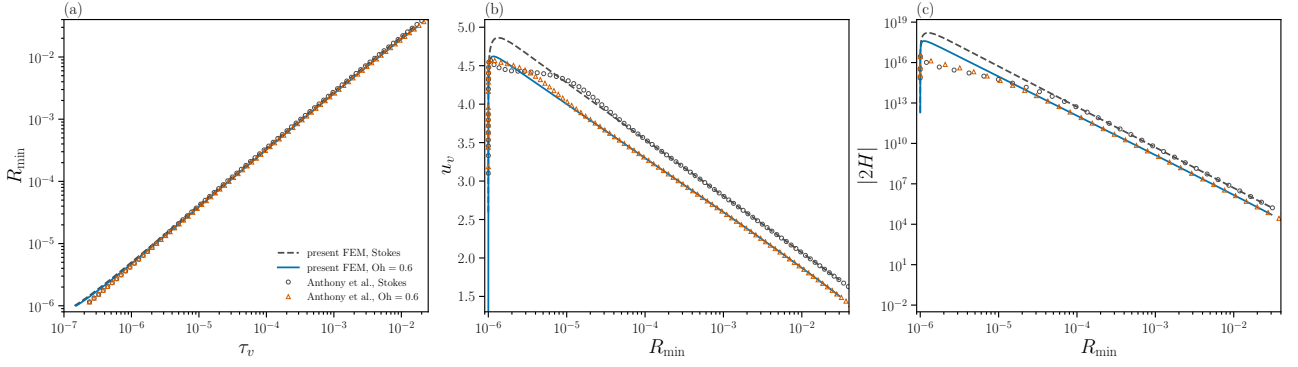


Figure 2: Both FEM curves and both digitised series from Anthony et al. [1], Fig. 3: (a) radius against viscous coalescence time, (b) velocity against radius and (c) magnitude of twice the mean curvature against radius. Colours and markers follow Figure 1. The power-law contact-time construction affects (a) only. No marker-fitted offsets are used.

The median differences $100(\text{FEM}/\text{Anthony} - 1)$ are listed below. The bands refer to τ_v for radius and $R_{\min}$ for velocity and curvature. Markers outside the computed coverage are excluded.

Quantity	Abscissa	10^{-5} – 10^{-4}	10^{-4} – 10^{-3}	10^{-3} – 10^{-2}	$\geq 10^{-2}$
$R_{\min}$	τ_v	+0.062%	−0.023%	−0.029%	−0.033%
u_v	$R_{\min}$	−0.133%	+0.005%	+0.028%	+0.044%
$ 2H $	$R_{\min}$	+3.19%	−3.50%	−3.04%	−3.37%

For $R_{\min} \geq 10^{-4}$, velocity differences range from −0.027% to +0.057%. Developed curvature is approximately 3–4% lower. The preceding curvature band is still transient: differences span −3.86% to +73.31%. The earliest finite-Oh velocity markers differ by up to about 3.8%; the radius comparison is also contact-time sensitive. We do not extend developed-range agreement to startup.

3 Verification and convergence

The inertial normal-mode test gives frequency errors of −0.044% and −0.019% at 200 and 400 steps per period. The $\text{Oh} = 10^{12}$ limit reproduces Stokes steps to 10^{-14} relative. The finite-Oh reference reaches $R_{\min} = 0.03037$ in 878 steps and 133 remeshes; its relative volume drift is 2.85×10^{-7} .

At $R_0 = 10^{-3}$ the frame checks differ by at most approximately 0.045%. The tip-map and half-time-step checks have 95th-percentile absolute differences below 0.003%, not uniform maximum errors. Bulk-grading 0.07 and remesh-growth 1.05 partners remain outstanding; we do not claim complete finite-Oh mesh and remeshing convergence.

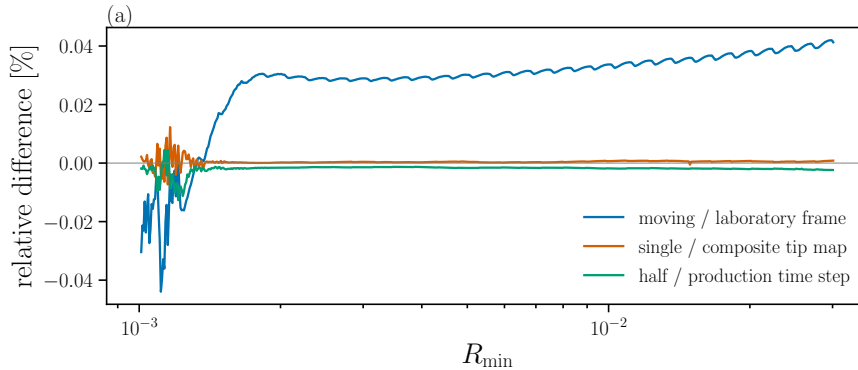


Figure 3: Completed finite-Oh frame, tip-map and time-step checks at $R_0 = 10^{-3}$, compared at equal radius from $1.01R_0$. Blue: moving versus laboratory frame; orange: single versus composite tip map; green: half versus production time step on the same single map. Bulk-grading and remeshing partners remain outstanding.

4 Independent Stokes BIM

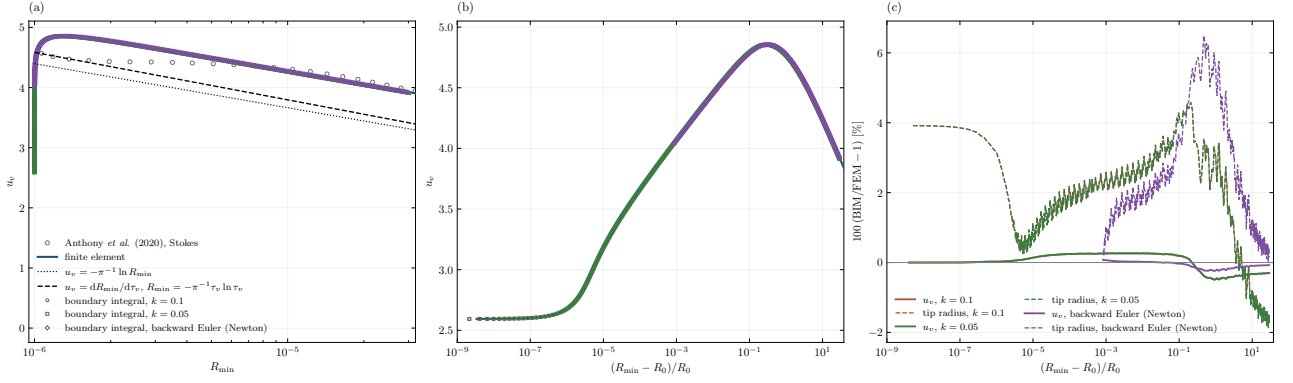


Figure 4: Independent Stokes BIM, Stokes FEM and Anthony’s Stokes startup for $R_0 = 10^{-6}$. The backward-Euler Newton series is within 0.243% of FEM; the linearly implicit grading series are within approximately 0.489%. Velocity is the discriminating observable; tip-curvature estimators carry approximately 4% uncertainty. This comparison contains no independent inertial BIM calculation.

The independently implemented FEM and BIM agree through the Stokes startup peak for the same initial bridge. The BIM solves a Stokes integral equation on a spline meridian; it does not borrow FEM fields or pressure data. The backward-Euler Newton series differs from FEM by at most 0.243%, and the two linearly implicit grading series by at most approximately 0.489%. The gradings agree much more closely with each other. This is code-to-code verification of Stokes flow, not a second validation of finite inertia. There is no additional independent finite-Oh boundary solver in the present comparison. The cause of the difference from Anthony’s Stokes startup is not established here.

5 Initial-radius independence

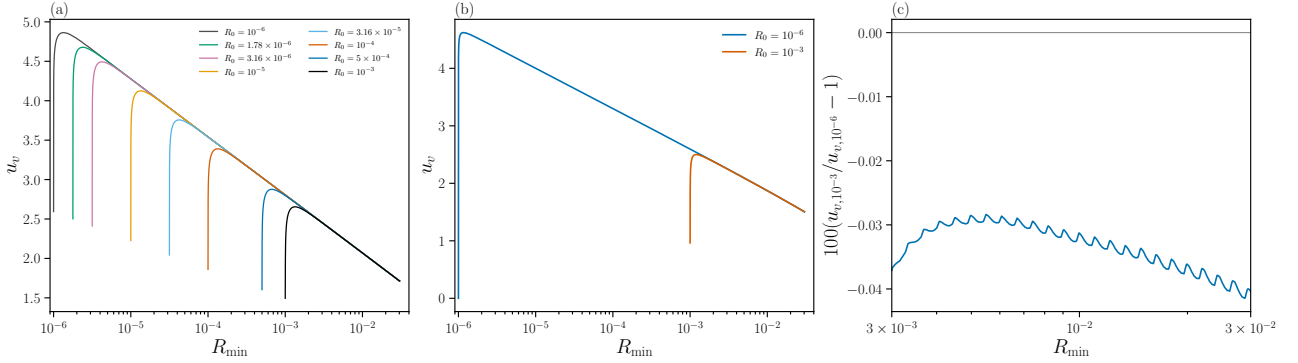


Figure 5: (a) Eight Stokes initial radii from 10^{-6} to 10^{-3} ; (b) the two finite-Oh endpoint radii; (c) their velocity difference at equal radius. At $\text{Oh} = 0.6$, the histories agree within 0.042% for $R_{\min} \geq 3 \times 10^{-3}$, without offsets. The finite-Oh intermediate-radius family is not yet complete.

The eight Stokes histories join the smallest bridge’s curve within 0.1% by approximately 2.3 – $2.7R_0$. At $\text{Oh} = 0.6$, the two endpoint radii $R_0 = 10^{-3}$ and 10^{-6} agree to within 0.042% for $R_{\min} \geq 3 \times 10^{-3}$. Intermediate finite-Oh radii have not yet been computed. This two-radius check is distinct from the completed Stokes family.

6 Finite inertia relative to Stokes

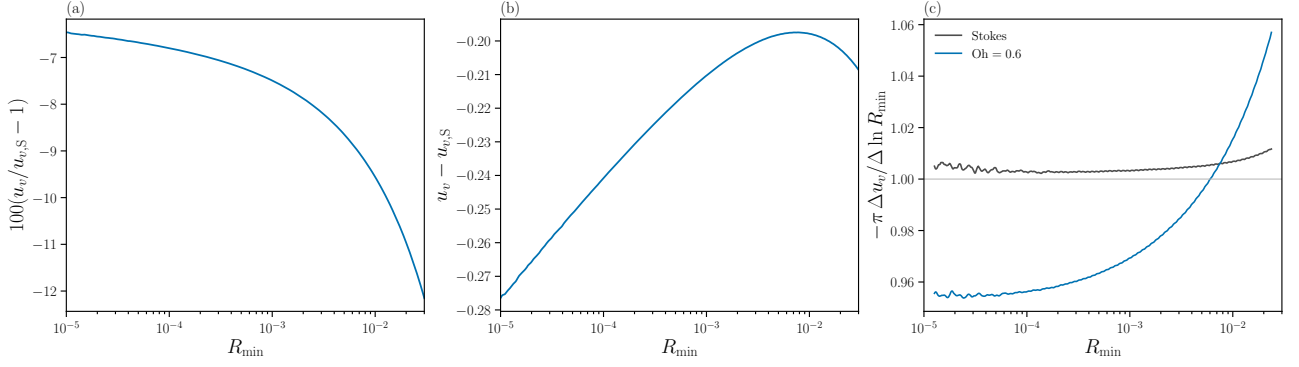


Figure 6: Finite-Oh velocity deficit at equal radius: (a) relative, (b) absolute and (c) local logarithmic slopes normalized by $1/\pi$, measured using secants across 0.2 decades. Grey: Stokes; blue: Oh = 0.6. The relative deficit grows from 6.46% at $R_{\min} = 10^{-5}$ to 12.15% at 0.03.

At equal neck radius, the finite-Oh velocity deficit is

$R_{\min}$	$R_{\min}/\text{Oh}$	$u_v - u_{v,S}$	Relative difference
10^{-5}	1.67×10^{-5}	-0.2765	-6.46%
10^{-4}	1.67×10^{-4}	-0.2408	-6.80%
10^{-3}	1.67×10^{-3}	-0.2103	-7.49%
10^{-2}	1.67×10^{-2}	-0.1979	-9.56%
0.03	0.05	-0.2086	-12.15%

The absolute deficit remains approximately -0.28 to -0.20 ; its relative magnitude grows because the Stokes velocity falls. Writing $u_v = \pi^{-1} \ln(C/R_{\min})$ defines the diagnostic $C_{\text{eff}} = R_{\min} \exp(\pi u_v)$. The inferred finite-Oh/Stokes ratio is 0.4196 at $R_{\min} = 10^{-5}$, 0.5371 at 10^{-2} and 0.5193 at 0.03. The first-decade median is approximately 0.444. The finite-Oh logarithmic slope, normalized by $1/\pi$, has decade medians of approximately 0.955, 0.961 and 0.985 between 10^{-5} and 10^{-2} .

A changed outer cut-off of the Stokes logarithm is therefore plausible, but a single constant shift cannot describe the weak slope change. The present history reaches only $R_{\min}/\text{Oh} = 0.05$ and does not test the crossover at $R_c \approx \text{Oh}$. An Oh ladder reaching that scale is needed to distinguish the cut-off interpretation from other finite-inertia corrections.

7 Reproducibility

The accompanying repository page records the build commands, registered run lists and source checksums. A compact plot-input bundle permits reproduction without raw meshes or field histories. The scripts retain the original visco-capillary time, exclude extrapolation and emit the comparison statistics separately from the figures. All six figure environments have reusable TeX sources; graphics use embedded fonts.

References

- [1] Christopher R. Anthony, Michael T. Harris, and Osman A. Basaran. Initial regime of drop coalescence. *Physical Review Fluids*, 5:033608, 2020. doi: 10.1103/PhysRevFluids.5.033608.